

UPI FRAUD DETECTION SYSTEM

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**A Mini Project Report
Submitted for the Subject
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AI-BASED FRAUD DETECTION SYSTEM

1. Title

UPI Fraud Detection System Using Machine Learning

2. Theoretical Background

UPI has made digital payments quick and convenient, but the growing number of transactions has also increased the risk of fraudulent activities. Machine Learning can help detect such fraud by learning patterns from historical transaction data.

This project uses a Random Forest Classifier, which combines multiple decision trees to classify transactions as genuine or fraudulent. Transaction data is preprocessed using feature engineering, missing-value handling, and One-Hot Encoding before being given to the model.

The system considers factors such as transaction amount, time of transaction, transaction type, merchant category, payment source, approval method, suspicious UPI-handle patterns, urgent wording, and screen-sharing requests. The model generates a fraud-risk score, which is categorized into Low, Medium, or High Risk to help assess the safety of a transaction.

The project also evaluates the model using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix, while checking for possible data leakage that could lead to misleadingly high performance.

3. Introduction

The rapid adoption of **UPI (Unified Payments Interface)** has transformed the way people make digital payments by providing fast and convenient transactions. However, the increasing use of UPI has also led to a rise in fraudulent activities such as suspicious payment requests, fake or lookalike UPI handles, and social-engineering-based scams.

The UPI Fraud Detection System is a **Machine Learning-based solution** developed to identify potentially fraudulent transactions by analyzing important transaction and payment-related factors. It uses a Random Forest Classifier trained on historical UPI transaction data to identify patterns associated with fraud.

The system analyzes factors such as transaction amount, transaction type, time, merchant category, payment source, approval method, UPI-handle patterns, urgent requests, and screen-sharing requests. Based on these factors, it **generates a Fraud Risk Score** and **categorizes the transaction into Low, Medium, or High Risk**, helping users make more informed decisions before proceeding with a payment. The main objective of the project is to provide an automated

and efficient method for fraud detection. Such a system can reduce manual effort, improve detection speed, and assist organizations in identifying transactions that require further investigation.

4. Literature Survey

Various studies have explored the use of **Machine Learning for financial and digital payment fraud detection**. Traditional rule-based systems rely on predefined conditions, while Machine Learning approaches can learn fraud patterns from historical transaction data.

Research on fraud detection has commonly used algorithms such as **Decision Trees, Logistic Regression, Support Vector Machines, Random Forest, and ensemble methods**. Among these, Random Forest is widely suitable for structured transaction data because it can handle multiple features and capture complex relationships.

Previous studies also highlight challenges such as **imbalanced datasets, changing fraud patterns, and data leakage**. Therefore, reliable fraud detection requires proper feature selection, preprocessing, and evaluation using metrics such as **Precision, Recall, F1-Score, and Confusion Matrix**.

Based on these concepts, the proposed project applies a **Random Forest-based approach** to UPI transaction data and considers practical fraud indicators such as transaction amount, transaction type, payment context, suspicious UPI handles, urgent requests, and screen-sharing requests.

5. Methodology Used

The methodology of the UPI Fraud Detection System follows a systematic Machine Learning workflow, starting from transaction data preparation and ending with fraud-risk prediction.

1. **Data Collection:** A UPI transaction dataset containing 26,393 records and 65 columns is used. The target variable is `_fraud` indicates whether a transaction is genuine or fraudulent.
2. **Data Preprocessing:** The dataset is examined for missing values and relevant transaction attributes are selected. Numerical and categorical data are prepared separately for further processing.
3. **Feature Engineering:** Raw transaction information is converted into useful model features. These include transaction amount, hour of transaction, urgent-language indication, screen-sharing request, transaction type, merchant category, session source, authorization method, lookalike/misspelled UPI handle, and recently created UPI handle.
4. **Feature Encoding:** Categorical features are converted into numerical representations using One-Hot Encoding, while missing numerical values are handled using appropriate imputation.

5. **Train-Test Split:** The prepared dataset is divided into 80% training data and 20% testing data using stratified sampling so that both classes are represented appropriately.
6. **Model Training:** A Random Forest Classifier with 300 decision trees is trained on the selected features. The model learns patterns that distinguish fraudulent transactions from genuine transactions.
7. **Model Evaluation:** The trained model is evaluated using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix to measure its classification performance.
8. **Risk Prediction:** For a new transaction, the model calculates the probability of fraud. This probability is converted into a Fraud Risk Score from 0–100 and classified into Low, Medium, or High Risk.
9. **Data Leakage Analysis:** Since an unusually perfect model performance can indicate leakage, the dataset is additionally examined for features that may unintentionally reveal the fraud label. This helps in understanding whether the model's performance is realistically reliable.

6. Current Trends

The use of **Artificial Intelligence and Machine Learning** in digital payment security is increasing, especially for identifying suspicious transaction patterns. Modern fraud detection systems focus on **real-time monitoring, behavioural analysis, and automated risk assessment** rather than relying only on fixed rules. Factors such as unusual transaction amounts, payment behaviour, suspicious requests, and abnormal activity can be analysed to detect potential fraud.

Another important trend is the use of **risk scoring and explainable AI**, which helps classify transactions according to their level of risk and provides better understanding of why a transaction may be suspicious. Since fraud techniques continuously evolve, modern systems also focus on **adaptive models, continuous monitoring, and regular model improvement** to maintain effective fraud detection.

7. Conclusion

The **UPI Fraud Detection System** successfully demonstrates how Machine Learning can be applied to identify and assess potentially fraudulent digital payment transactions. The project follows a complete workflow, **including data preprocessing, feature engineering, feature encoding, model training, evaluation, and fraud-risk prediction**. The Random Forest Classifier analyses important transaction-related factors such as amount, transaction type, payment source, authorization method, suspicious UPI-handle patterns, urgent requests, and screen-sharing requests to determine the likelihood of fraud.

The system converts the model's fraud probability into a **Fraud Risk Score** and categorizes transactions as **Low, Medium, or High Risk**, making the prediction easier to understand and act upon. Model performance is evaluated using **Accuracy, Precision, Recall, F1-Score, and Confusion Matrix**, while data-leakage analysis highlights the importance of validating unusually high performance rather than assuming that a perfect score always represents a reliable model.

Overall, the project provides a practical foundation for Machine Learning-based UPI fraud detection and demonstrates the importance of combining data analysis, appropriate feature selection, model evaluation, and user-oriented risk assessment. Although the system is developed for academic purposes and cannot replace real banking security infrastructure, it can be further enhanced with larger real-world datasets, real-time monitoring, advanced algorithms, explainable AI, and continuously updated fraud patterns to develop a more robust fraud detection solution.

8. Acknowledgement

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